One Joint Transform for the Complete Exponential Cramér–Lundberg Ruin Atlas

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Abstract

For the classical compound-Poisson surplus with exponential claims, we derive one closed joint transform of ruin time and deficit. It gives ultimate ruin on both sides of the safety-loading wall and proves conditional overshoot independence. No periodic-orbit or arithmetic target conclusion is drawn.

1 Frozen process and joint penalty

Let N_t be Poisson with rate $\nu \geq 0$, let Y_i be independent $\mathrm{Exp}(\beta)$ claims with $\beta > 0$, and fix $c > 0$, $u \geq 0$. The surplus, strict ruin time, and deficit are

$$U_t = u + ct - \sum_{i=1}^{N_t} Y_i, \qquad \tau = \inf\{t \geq 0 : U_t < 0\}, \qquad D = -U_\tau. \tag{1}$$

Thus $u = 0$ is not initially ruined. The Route-A dynamical owner is the killed process

$$X_t = U_t \quad (t < \tau), \qquad X_t = \Delta \quad (t \geq \tau),$$

whereas the joint transform below is a first-passage functional of the underlying surplus U. Gerber and Shiu [1] are cited only for discounted-penalty lineage; Drekic and Willmot [2] are cited only for exponential-claim ruin-time lineage. No formula or proof below is outsourced.

For $q, s \geq 0$ put

$$\Phi_{q,s}(u) = \mathbb{E}_u[e^{-q\tau - sD}; \tau < \infty]$$

and define

$$r_q = \frac{c\beta - \nu - q + \sqrt{(c\beta - \nu - q)^2 + 4c\beta q}}{2c}. \tag{2}$$

Theorem 1 (Complete joint transform)**.** *For every frozen parameter,*

$$\boxed{\Phi_{q,s}(u) = \frac{\beta - r_q}{\beta + s} e^{-r_q u}.} \tag{3}$$

Here $0 \leq r_q \leq \beta$. If $\nu = 0$, then $r_q = \beta$ and (3) is zero. If $\nu > 0$, conditional on ruin, $D \sim \mathrm{Exp}(\beta)$ and D is independent of τ.

Proof. First-jump conditioning gives, for $u > 0$,

$$\begin{aligned} 0 = c\Phi'(u) - (\nu + q)\Phi(u) + \nu \int_0^u \Phi(u - y)\beta e^{-\beta y}\, dy \\ + \nu \int_u^\infty e^{-s(y-u)} \beta e^{-\beta y}\, dy. \end{aligned} \tag{4}$$

The last integral is $\beta e^{-\beta u}/(\beta + s)$. To prove that an exponential ansatz is exhaustive, introduce the convolution state

$$J(u) = \int_0^u \Phi(u - y)\beta e^{-\beta y}\, dy.$$

Equation (4) is equivalent to the two-dimensional inhomogeneous system

$$\begin{aligned} c\Phi' &= (\nu+q)\Phi - \nu J - \frac{\nu\beta}{\beta+s}e^{-\beta u}, \\ J' &= \beta(\Phi - J), \qquad J(0)=0. \end{aligned} \tag{5}$$

Its homogeneous mode e^{-ru} exists exactly when

$$cr^2 - (c\beta - \nu - q)r - q\beta = 0, \tag{6}$$

so these two modes exhaust the homogeneous solution space (with the usual generalized mode at a double root). A particular solution of (5) is $(\Phi, J) = (0, -\beta e^{-\beta u}/(\beta+s))$. For $q > 0$, the two roots have opposite signs, so boundedness removes the negative root and retains (2). At $q = 0$ with $0 < \nu < c\beta$, both 0 and $R = \beta - \nu/c$ give bounded nonnegative local solutions. Here the strong law for $Z_t = \sum_{i \le N_t} Y_i - ct$ gives $Z_t/t \to \nu/\beta - c < 0$, hence $M = \sup_{t\ge 0} Z_t < \infty$ almost surely and $\Phi_{0,s}(u) \le \mathbb{P}(M > u) \to 0$; this removes the constant mode and selects R. If $\nu > c\beta$, boundedness removes the growing negative-root mode and retains $r = 0$. At equality the second generalized mode grows linearly and is again removed by boundedness. For the surviving mode, $J = A\beta e^{-ru}/(\beta - r) - \beta e^{-\beta u}/(\beta + s)$, so $J(0) = 0$ forces $A = (\beta - r)/(\beta + s)$. This mode exhaustion plus the stated chamber boundaries proves uniqueness, rather than assuming it from the ansatz. If $\nu = 0$ there are no claims and $\Phi \equiv 0$, which is (3) with $r_q = \beta$. Continuity under the common-path coupling extends the solution from $u > 0$ to $u = 0$; equality at a claim has probability zero, so the strict-passage convention is preserved.

For $\nu > 0$, (3) factors as $\Phi_{q,s} = \{\beta/(\beta+s)\}\Phi_{q,0}$. This identifies the conditional Laplace transform of D and its independence from τ. Pathwise, the same fact is exponential memorylessness beyond the pre-ruin reserve. □

2 Evidence and Route-A boundary

The canonical receipt contains 36 exact regime rows, 448 joint-transform rows, 144 conditional-first-mean rows, 12 martingale/supremum rows, and six boundary rows. An independent checker passes 4,487 assertions; SymPy passes 15 identities; fresh replay is byte-exact; and 26/26 repaired-hash or stale-hash mutations are rejected. Finite rows are regression checks, not proof of the continuum theorem.

The Route-A tuple is

$$(\text{A0\_FAIL}, \text{A1\_FAIL}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FAIL}), \qquad \text{ROUTE\_A\_REJECTED}. \tag{10}$$

There is no arithmetic origin. More precisely, the killed PDMP/Markov semigroup has no intrinsic deterministic, enumerable primitive-periodic-orbit owner. No rational-prime carrier, logarithmic clock, target determinant, target zero match, Hilbert–Pólya operator, or Route-B input follows. Scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

References

[1] H. U. Gerber and E. S. W. Shiu, “On the time value of ruin,” *North American Actuarial Journal* **2** (1), 48–72 (1998), doi:10.1080/10920277.1998.10595671.

[2] S. Drekic and G. E. Willmot, “On the density and moments of the time of ruin with exponential claims,” *ASTIN Bulletin* **33** (1), 11–21 (2003), doi:10.2143/AST.33.1.1036.